LUNDS TEKNISKA HÖGSKOLA
MATEMATIK

TENTAMENSSKRIVNING
Matristeori
4-14 December 2024

Hand in solutions to at most 7 of the 8 problems below if you follow any of the 7.5hp courses (FMAN71, MATC70 and FMA122F) or 6 problems if you follow any of the 6hp courses (FMAN70, FMA121F). Credits can be given for partially solved problems. Write your solutions (in English or Swedish) neatly and explain your calculations.

The exam should be submitted through the canvas page by August 29. **Write your name, section-year, and the code for the course you are following on the first page. If you have preferences for when to take the oral exam, also write this on the first page.** The solutions should be submitted as a pdf-file. We encourage you to hand in scanned hand-written solutions, but make sure that the solutions are easily readable after scanning. Programs that you have written (e.g. matlab or maple files) should also be submitted through the canvas page. Make sure to reference these and explain how you have used them in your main solution-pdf.

You may use books and computer programs (e.g. Matlab and Maple), but it is not permitted to get help from other persons (or chat bots). You do not need to explain the elementary matrix operation such as matrix multiplications made by the computer, but should explain steps in more complicated calculations such as jordanization. Ask if you are not sure!

You can use any theorem in the book without proof but not exercises without explaining the solutions.

All matrices below are complex matrices if nothing else is specified explicitly.

Good luck!

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1. Let $\mathcal{P}$ be the set of polynomials of degree at most 2. Show that $\mathcal{L} : p(x) \mapsto p'(x)+p(x)$ is a linear operator on $\mathcal{P}$. Determine the matrix representation of this operator in the basis $1, x, x^2$ and the Jordan form. Is there a polynomial in $\mathcal{P}$ that is mapped to itself by $\mathcal{L}$? Can $\mathcal{L}$ be diagonalized?

2. Let $f : \mathbb{C} \to \mathbb{C}$ be given by $f(q) = -\frac{p}{2} + \sqrt{\frac{p^2}{4} - q}$, for some number p. Show that if f is defined on the square matrix Q then $X = f(Q)$ gives a solution to the matrix equation $X^2 + pX + Q = 0$. If

$$B = \begin{pmatrix} -10 & -24 & 12 \\ 3 & 8 & -3 \\ -6 & -12 & 8 \end{pmatrix},$$

determine if B is diagonalizable and find a solution A to $A^2 + 2A + B = 0$. Is the solution unique?

3. Let A be a square matrix of size $n \times n$ for which $a_{ij} = 1$ when $i-j \neq 1$ and $a_{i,i-1} = -n$ for $i = 2, 3 \ldots n$. Compute $\det A$. Is A invertible?

4.

a) Let $p(x) = x^3 - 7x + 6$. Show that any matrix A that fulfills $p(A) = 0$ is diagonalizable.

b) Suppose that the matrix H has left inverse L, that is $LH = I$. Show that the matrix HL is diagonalizable.

5. Show that the matrix $\begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$ is congruent with $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Suppose that A is a real $m \times n$ matrix with $m = \operatorname{rank}(A) < n$. Determine the signature of the of the quadratic form given by $f(u, v) = u^T A^T A v$, and characterize it as either positive semi-definite, indefinite or negative semi-definite.

6. Show that $(x, y)_M := x^H M y$ is an inner product on $\mathbb{C}^n$ if and only if the hermitian $n \times n$ matrix M is positive definite.

Furthermore, if M is positive definite show that any invertible matrix A has a unique decomposition $A = PR$, where $P^H M P = I$ and R is upper triangular with real positive diagonal elements.

7. Let A, B be two square matrices such that $AB - BA = A$.

a) Show that $A^k B - BA^k = kA^k$ for each $k = 1, 2, \ldots$

b) Show that A is a nilpotent matrix.

8. Suppose that A is an $n \times n$ Hermitian matrix with eigenvalues $\lambda_1 \leq \lambda_2 \leq ... \leq \lambda_n$ and corresponding orthonormal basis of eigenvectors $u_1, u_2, ..., u_n$.

a) Let $\mathcal{V}$ and $\mathcal{H}$ be linear subspaces of $\mathbb{C}^n$ with $\dim \mathcal{V} = k$ and $\dim \mathcal{H} = n - k + 1$. Show that $\mathcal{V} \cap \langle u_k, ..., u_n \rangle \neq \{0\}$ and $\mathcal{H} \cap \langle u_1, ..., u_k \rangle \neq \{0\}$.

b) Show that for any linear subspaces $\mathcal{V}$ and $\mathcal{H}$ with $\dim \mathcal{V} = k$ and $\dim \mathcal{H} = n-k+1$ we have

$$\min_{0 \neq x \in \mathcal{H}} \frac{x^H A x}{x^H x} \leq \lambda_k \leq \max_{0 \neq x \in \mathcal{V}} \frac{x^H A x}{x^H x}.$$

Furthermore show that equality holds if $\mathcal{V} = \langle u_1, ..., u_k \rangle$ and $\mathcal{H} = \langle u_k, ..., u_n \rangle$.

c) If $\tilde{A}$ is the $(n-1) \times (n-1)$ matrix obtained when removing the last row and column from A and $\eta_1 \leq \eta_2 \leq ... \leq \eta_{n-1}$ are the eigenvalues of $\tilde{A}$ show that

$$\lambda_k \leq \eta_k \leq \lambda_{k+1}.$$